4.

(2) 令 $\dot{I}_2 = 0$

$$A = \frac{\dot{U}_1}{\dot{U}_2} = -1$$

$$C = \frac{\dot{I}_1}{\dot{U}_2} = 0$$

令 $\dot{U}_2 = 0$

$$B = \frac{\dot{U}_1}{-\dot{I}_2} = 0$$

$$D = \frac{\dot{I}_1}{-\dot{I}_2} = -1$$

$$T = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

2.

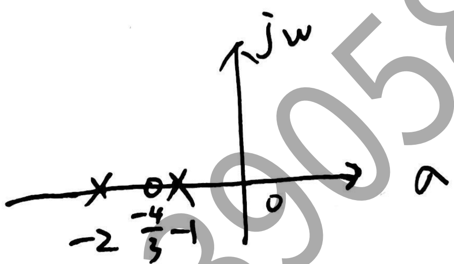
$$H(s)=\frac{1}{s+1}+\frac{2}{s+2}$$

$$=\frac{s+2+2(s+1)}{(s+1)(s+2)}$$

$$=\frac{3s+4}{(s+1)(s+2)}$$

令 $3s+4=0$ 零点：$-\frac{4}{3}$

令 $(s+1)(s+2)=0$ 极点：$-1, -2$



1.

$$U_c(0+) = U_c(0-) = U_s =$$

$$U_c(\infty) = I_s R_2 = 2 \times 10 = 20V$$

$$R_0 = R_2 = 10\Omega$$

$$\tau = R_0 C = 10 \times 0.1 \times 10^{-6} = 1 \times 10^{-6}(s)$$

$$U_c(t) = U_c(\infty) + [U_c(0+) - U_c(\infty)]e^{-\frac{t}{\tau}}$$

$$= 20 + (U_s - 20)e^{-10^6 t}\ (V)\ (t > 0)$$

1.

$$U_c(0_+) = U_c(0_-) = U_s = 12V$$

$$U_c(\infty) = I_s R_2 = 2\times 10 = 20V$$

$$R_0 = R_2 = 10\Omega$$

$$\tau = R_0 C = 10\times 0.1\times 10^{-6} = 1\times 10^{-6}(s)$$

$$U_c(t) = U_c(\infty) + [U_c(0_+) - U_c(\infty)]e^{-\frac{t}{\tau}}$$

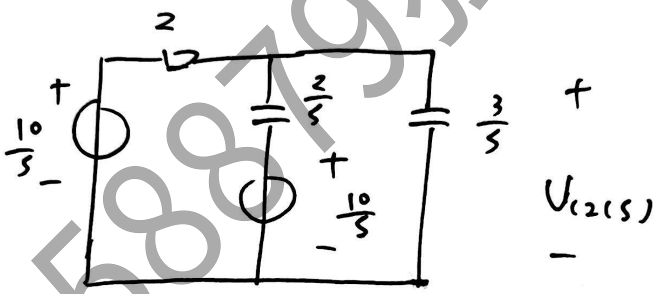
$$= \cancel{20 + (U_s - 20)e^{-10^6 t}\,(V)\ (t>0)}$$

$$= 20 - 8e^{-10^6 t}\,(V)\ (t>0)$$

QQ13139058879独家发行

5.

$U_{C1}(0-) = 10V$



$$U_{C2}(s) = \frac{\frac{\frac{10}{s}}{2} + \frac{\frac{10}{s}}{\frac{2}{s}}}{\frac{1}{2} + \frac{s}{2} + \frac{s}{3}} = \frac{5s+5}{\frac{5}{6}s(s+\frac{3}{5})}$$

$$= \frac{10}{s} + \frac{-4}{s+0.6}$$

$$U_{C2}(t) = 10 - 4e^{-0.6t} \, (V)$$

2.

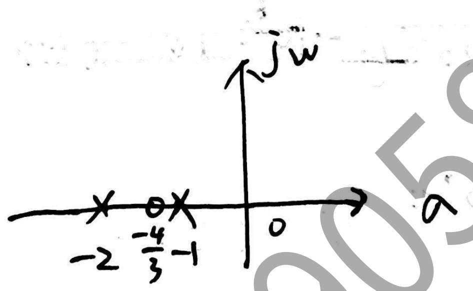
$$H(s)=\frac{1}{s+1}+\frac{2}{s+2}$$

$$=\frac{s+2+2(s+1)}{(s+1)(s+2)}$$

$$=\frac{3s+4}{(s+1)(s+2)}$$

令 $3s+4=0$ 零点：$-\frac{4}{3}$

令 $(s+1)(s+2)=0$ 极点：$-1, -2$.



4.

设 $T = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$

(1) 令 $\dot{I}_2 = 0$

$$A = \frac{\dot{U}_1}{\dot{U}_2} = 1$$

$$C = \frac{\dot{I}_1}{\dot{U}_2} = 0$$

令 $\dot{U}_2 = 0$

$$B = \frac{\dot{U}_1}{-\dot{I}_2} = 0$$

$$D = \frac{\dot{I}_1}{-\dot{I}_2} = 1$$

$$T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

3.

$$|Z| = \sqrt{12^2+16^2} = 20\Omega$$

$$\cos\varphi = \frac{12}{20} = 0.6$$

$$U_p = \frac{U_L}{\sqrt{3}} = 220V$$

$$I_p = \frac{U_p}{|Z|} = \frac{220}{20} = 11A$$

$$I_L = I_p = 11A$$

$$P = \sqrt{3}\,U_L I_L \cos\varphi$$

$$= \sqrt{3}\times 380\times 11\times 0.6 = 4344W$$

6. $u_L = L\frac{di_L}{dt}$

$$i_c = c\frac{du_c}{dt}$$

$$i_R = \frac{u_c}{R}$$

$$i_L = c\frac{du_c}{dt} + \frac{u_c}{R}$$

$$e(t) = L\frac{di_L}{dt} + u_c$$

$$\Rightarrow \begin{cases} \frac{du_c}{dt} = -\frac{1}{Rc}u_c + \frac{1}{c}i_L \\ \frac{di_L}{dt} = -\frac{1}{L}u_c + \frac{1}{L}e(t) \end{cases}$$

$$\begin{bmatrix} \frac{du_c}{dt} \\ \frac{di_L}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{1}{Rc} & \frac{1}{c} \\ -\frac{1}{L} & 0 \end{bmatrix}\begin{bmatrix} u_c \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix}e(t)$$